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TAFILA TECHNICAL UNIVERSITY
Faculty of Engineering
Department of Computer and Communications Engineering

Microprocessor and Assembly Language/ Microprocessor 1
First Exam

Student Name: [redacted] Time allowed: 60 minutes

18/25

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Question#1
Select the correct answer for each of the following (**6 pts**):

[1] Which instruction among the following is Valid?

a) PUSH BL
b) ADD BL, BH
c) MOV BL, 10101010B
d) MOV BL, CX

[2] How many address bits are required to address/index **2Mbytes** of memory?

a) 20 b) 21 c) 22 d) 24

[3] What is the address of the last memory location in a 2Kbytes memory chip?

a) 1sFF b) 3FF c) 7FF d) FFF

[4] Splitting the internal structure of the 8086 microprocessor into two sections (the **BIU** and the **EU**) makes the CPU process information faster by:

a) Verifying the pipelining concept
b) Increasing the address bus
c) Performing two loads from memory at the same time
d) Increasing the data bus

[5] Which one among the 8086 microprocessor resources is used to store fetched instructions?

a) Registers b) Instruction Decoder
c) ALU d) Instruction Queue

[6] What is the size of the data bus in the 8088 microprocessor?

a) 8 b) 16 c) 24 d) 32

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Question#2

Given the following 8086 assembly and machine code which is stored in the code segment, answer the followed questions? **(7 pts)**

Logical Address	Machine Code	Assembly Instructions
54B0:0346	B078	MOV AL, 78H
54B0:0348	B426	MOV AH, 26H
54B0:034A	02E0	ADD AH, AL

[1] Determine the physical addresses of the two memory locations that hold the third instruction?

54B00 + 034A ⟹ 54B00
= 54E4A ✓ 1
034A
54E4A

[2] What is the value of IP register needed to fetch the second instruction?

0348 ✓ 1

[3] What would be the content of AX register after executing the whole code?

AL = 78H
AH = 26H
AH = AH + AL

AH = 26 + 78 ⟹ 26
78
9E
AH = 9E
AX = 9E 78 ✓ 2

[4] Determine the physical addresses of the last two locations in Code Segment?

upper ⟹ 54B00 + FFFF ⟹ 54B00
= 64AFF ✓ FFFF
64AFF

before upper ⟹ 54B00 + FFFE ⟹ 54B00
= 64AFE ✓ 2 FFFE
64AFE

↳ or from upper we will know the before

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Question#3

Given that the register values of an 8086 microprocessor are: **(6 pts)**

CS : 43A0H	SP : 0B2BH
DS : 22A0H	BP : 0269H
SS : 6F20H	DI : 4598H
SI : 02BCH	

[1] Indicate the PHYSICAL addresses accessed by the following instructions, if you believe the instruction is invalid for any reason indicate that there is an ERROR in the instruction. Assume that all instructions take place with the above register contents, and the actions of instruction A will not affect the registers seen by subsequent instructions.

a) PUSH AX

SS:SP ⟹ 6F20 : 0B2B

push 1 ⟹ 6F200 + 0B2B = ⟹ 6F200 + 0B2B = 6FD2B

phy 1 ⟹ = 6FD2B

SS:SP-1 ⟹ 6F20 : 0B2A

push 2 ⟹ 6F200 + 0B2A = ⟹ 6F200 + 0B2A = 6FD2A

phy 2 ⟹ = 6FD2A

b) MOV AL, [200]

↳ ERROR

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[2] Determine the value of the BX register that loads the content of memory location 23268H using the instruction MOV AL, [BX]?

```
 2
23268
-22A00
 0868
```

phys = DS x 10 + offset of [BX]

23268 = 22A00 + offset of [BX]

offset of [BX] = 23268 - 22A00

offeso f [BX] = 0868

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Tafila Technical University
Faculty of Engineering
Communications and Computer Engineering Department

Microprocessor and Assembly Language
Fall 2022 – 2nd Exam

Name: [redacted] **Time Allowed: 60 minutes**

Question#1

Choose the correct answer for each of the following (**9 points**):

1] Which register value is true after executing the following three instructions?

```
MOV AX, 21
MOV BH, 5
DIV BH
```

a) AL= 01 b) AH= 04 c) AX= 0104H d) AX= 0004

2] What would be the content of memory location 103 after defining the following?

```
ORG 100H
DATA1 DQ 1357924 68H
```

a) 35H b) 46H c) 92H d) 79H

3] What is the content of the AL register after executing the following?

```
MOV AL, 2AH
MOV CL, 3
SHR AL, CL
```

a) 51H b) 45H c) 50H d) 05H

4] The condition on which the instruction JBE takes place is:

a) CF= 0 and ZF= 0
b) CF= 1 and ZF= 1
c) CF= 1 or ZF= 0
d) CF= 1 or ZF= 1

5] Which instruction is used to convert ASCII to unpacked BCD?

a) ADD b) AND c) XOR d) CMP

6] Which instruction of the following use Register Addressing Mode?

a) MOV AL,29
b) MOV CX,[SI]
c) MOV AX,[200]
d) MOV CX,DX

7] Which instruction among the following would not change the content of register AL?

a) AND AL, 0FFH b) OR AL, 0FFH
c) XOR AL, 0FFH d) ADD AL, 0FFH

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8] What would be the value of CF, SF, and PF after executing the following?

MOV BH, 3Ch
ADD BH, 12h

a) CF= 0, SF= 0, PF=0
b) CF= 0, SF=0 , PF=1
c) CF= 1, SF=0 , PF=1
d) CF= 0, SF=1 , PF=0

9] Which of the following defines Packed BCD number equals 18?

a) n1 db '18'
b) n2 db 18
c) n3 db 18h
d) n4 dw 0108h

Question#2

Replace each one of the following with an equivalent instruction **(3 points)**:

1) NOT DH
   XOR DH, FF
2) MOV AL, 0
   SUB AL, AL
3) MOV DL, 2
   DIV DL
   MOV offset[DL], 2
   DIV offset[DL]

Question#3

Given the following 2-byte short-jump instruction **(6 points)**:

CS:IP	Instruction	Machine Code
2020H:1200H	JNZ Location	75XXH

a) Given ZF=0, What's the value of the XX that makes the code enters in an infinite-loop (stuck executing the JNZ Location instruction):

b) What's the value of XX that keeps the execution of the code sequentially regardless the value of the ZF?

XX = 00

c) If the value of XX is 32H, what makes the code executes the instruction located at address 2020H:1202H directly after the JNZ location instruction:

d) If the JNZ instruction transfers the execution of the code to physical address 21445H, then the value of XX is:

2020 : 1245

20200 + 1245 = 21445

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Question#4

a) Write a code to count the number of passing students? **(3 points)**

Grades db 40, 65, 82, 95, 72, 57, 62, 45, 35, 89
Count db ?

```
mov cl, 10
mov bl, 0
cmp al, 50
ja pass
mov count, bl
pass: offset Grades + 1
inc bl
dec cl
ja mark
```

b) Write an assembly program to count the numbers that are multiples of 3 in the following array? **(4 points)**:

X db 15, 20, 25, 36, 41, 12, 17
Count db ?

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